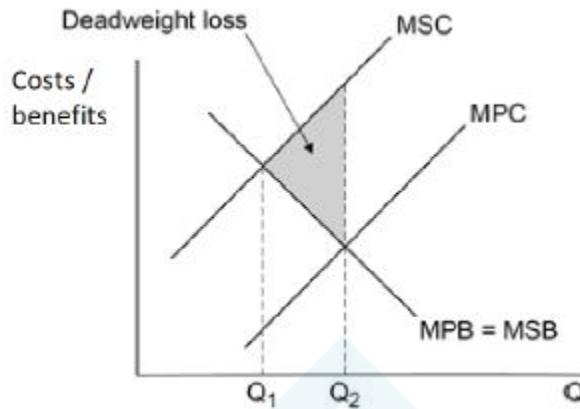


1.

Level of response	Response:	Max 9 marks
3	- is well organised and develops one or more of the key issues that are relevant to the question - shows sound knowledge and understanding of relevant economic terminology, concepts and principles - includes good application of relevant economic principles and/or good use of data to support the response - includes well-focused analysis with a clear, logical chain of reasoning - includes a relevant diagram that will, at the top of this level, be accurate and used appropriately	7 – 9 marks
2	- includes one or more issues that are relevant to the question - shows reasonable knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles and/or data to the question - includes some reasonable analysis but it might not be adequately developed and may be confused in places - may include a relevant diagram	4 – 6 marks
1	- is very brief and/or lacks coherence - shows some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - demonstrates very limited ability to apply relevant economic principles and/or data to the question - may include some very limited analysis but the analysis lacks focus and/or becomes confused - may include a relevant diagram but the diagram is not used and/or is inaccurate in some respects.	1 – 3 marks

Suggested diagram

An MSC/MSB externality diagram is expected, showing overproduction as a result of a negative externality in production. However, a demand and supply diagram illustrating overproduction resulting from a negative externality is also acceptable.



An externality diagram showing negative consumption externalities is also acceptable. The written response should relate appropriately to the chosen diagram.

Relevant issues include:

- definition of market failure
- explanation of diagram, showing over-production of goods due to the marginal external costs creating a divergence between marginal social cost and marginal private cost. The over-production of goods creates a deadweight loss
- explanation of why output in a free market will be where $MPC=MPB$, whereas the socially optimal output is where $MSC=MSB$
- explanation of economic (in)efficiency
- explanation of how firms have an incentive to make products short-lived, as such products are likely to be made with cheaper raw materials, less quality control and perhaps cheaper labour
- how designing products to fail may result in more sales in the future
- how the actions of producers are normally not constrained by any requirements to consider how products are disposed of at the end of their useful life
- how disposal of waste products via landfill or incineration imposes externalities on third parties
- analysis of what such externality impacts are and who pays for them
- how the rapid utilisation of scarce resources such as minerals, water, oil, accelerates the degradation of the environment and reduces the quantity available for the future
- how such environmental problems might be considered allocatively inefficient
- reference to extracts, including the fact that not only producers, but also transport companies and others benefit from the overproduction of goods.

[9]

2.

A

[1]

**3.**

For a relevant definition / description of property rights, e.g. a legally recognised right of ownership over a resource; or a definition of market failure.

For explaining how property rights can be lacking, e.g. on common land, on the high seas, in the atmosphere.

Pollution is an example of a negative externality. One failure is the pollution of a communal resource such as a piece of land, a lake or the atmosphere for any one example). Fresh air is a resource essential to life however the atmosphere can also be polluted by human activity. Because nobody owns the atmosphere the costs of pollution are external to the polluter and it is possible for an economic agent to pass these costs on to the wider community.

Another failure is the **depletion** of a communal resource, e.g. water reserves, fish stocks, wildlife, open spaces, and property rights incentivise conservation.

Make appropriate use of diagrams, e.g. an externality diagram.

[9]**4.**

D

[1]**6.****Areas for discussion include:**

- explaining: permit trading, quotas, environmental taxes, regulation, subsidising green technologies
- effects of emissions trading and alternatives, specifically on UK businesses (firms) in terms of:
 - costs
 - revenues
 - profits
- rationale for government intervention
- internalising externalities and “polluter pays” ideas
- command-and-control versus market-based solutions
- permit/carbon trading schemes in theory and practice
- alternatives and their implications compared and contrasted, e.g. quotas, taxes and subsidising green technologies
- efficacy/appropriateness of ‘offsets’
- possible conflicts between policies, e.g. environmental improvement versus growth and employment
- distortions, e.g. lenient treatment of favoured businesses/industries (mentioned in the data)
- the need for international action
- being ‘carbon neutral’ as a business objective

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the candidate’s response to the question.

[25]

**7.****Areas for discussion include:**

- definition of monopoly power
- explanation of how new monopolies have achieved market dominance
- explanation of how firms with monopoly power can exploit their power to restrict output and raise prices
- discussion of how the nature of monopoly is changing, as the rich world's demands shift away from heavy industry and towards information economies
- how monopoly regulation has largely been designed for a different context from the modern internet firms, and is (to some extent) trying to catch up with the pace of change
- how breaking up firms may lead to more competition and better customer outcomes, but could jeopardise the economies of scale enjoyed by large dominant firms and their customers
- how a tax would be likely to work: an indirect tax; 'super tax' on abnormal profits
- recognition that many services offered by these firms are free to users, paid for by third-party advertising, hence price-based interventions are unlikely to be effective
- one proposed policy in this area is to require tech firms to pay users for permission to collect and use personal data
- discussion of the international context. Many of the technology firms have significant operations in multiple countries, so any policy may need to be agreed/coordinated by a number of countries. Firms might be considered to be 'footloose' in terms of tax avoidance
- examples of regulation of such firms by eg EU Competition Commissioner's fine for Google
- recognition of how many concerns about the large Internet firms are about data protection and privacy rather than economic concerns
- overall assessment about whether there ought to be more intervention in this market, and whether it should be by breaking up the firms
- explanation of how the recent history has featured the natural demise of seemingly dominant firms in the information market (eg myspace, askjeeves, IBM) without any government intervention
- discussion of productive, allocative and dynamic efficiency
- market failure arguments
- government failure arguments.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response - includes well-focused analysis with clear, logical chains of reasoning - includes supported evaluation throughout the response and in a final conclusion.	21 – 25 marks
4	Sound, focused analysis and some supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response - includes some well-focused analysis with clear, logical chains of reasoning - includes some reasonable, supported evaluation.	16 – 20 marks
3	Some reasonable analysis but generally unsupported evaluation that: - focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response - includes some reasonable analysis but which might not be adequately developed or becomes confused in places - includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgments but these are not well-supported by arguments and/or data.	11 – 15 marks
2	A fairly weak response with some understanding that: - includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - includes some limited application of relevant economic principles to the given context and/or data to the question - includes some limited analysis but it may lack focus and/or become confused - includes attempted evaluation which is weak and unsupported.	6 – 10 marks
1	A very weak response that: - includes little relevant knowledge and understanding of economic terminology, concepts and principles - includes application to the given context which, at best, is very weak - includes attempted analysis which is weak and unsupported.	1 – 5 marks

[25]

**9.****Areas for discussion include:**

- definition of price discrimination
- the conditions necessary for price discrimination
- discussion of different types of price discrimination
- the disadvantages of price discrimination (higher price for many consumers, loss of consumer surplus, pricing seen as unfair eg for captive consumers such as rail season ticket holders)
- the advantages of price discrimination (lower prices for some customers with lower demand, typically those from low income households, customers with highest utility of a good or service likely to obtain the good, output likely to be higher than would be the case from a non-discriminating monopolist)
- price discrimination may be preferable to a single-price which results in secondary markets developing (eg ticket touts)
- price discrimination is good for the monopolist, increasing revenues and profits
- price discrimination is good for consumers who would otherwise have been unable to afford a good or service (eg last-minute holiday bookings) eases the impact of inequalities
- price discrimination may allow for cross-subsidisation of unprofitable goods or services
- the negative image of price discrimination, especially relating to accusations of exploitation levelled at monopolists in response to natural disasters
- discussion of the alternative to a discriminating monopolist (likely to be a non-discriminating monopolist rather than a competitive market)
- use of examples
- market failure arguments.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

Level of response	Response	Max 25 marks
5	Sound, focused analysis and well-supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response - includes well-focused analysis with clear, logical chains of reasoning - includes supported evaluation throughout the response and in a final conclusion.	21 – 25 marks
4	Sound, focused analysis and some supported evaluation that: - is well organised, showing sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors - includes some good application of relevant economic principles to the given context and, where appropriate, some good use of data to support the response - includes some well-focused analysis with clear, logical chains of reasoning - includes some reasonable, supported evaluation.	16 – 20 marks
3	Some reasonable analysis but generally unsupported evaluation that: - focuses on issues that are relevant to the question, showing satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present - includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response - includes some reasonable analysis but which might not be adequately developed or becomes confused in places - includes fairly superficial evaluation; there is likely to be some attempt to make relevant judgments but these are not well-supported by arguments and/or data.	11 – 15 marks
2	A fairly weak response with some understanding that: - includes some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely - includes some limited application of relevant economic principles to the given context and/or data to the question - includes some limited analysis but it may lack focus and/or become confused - includes attempted evaluation which is weak and unsupported.	6 – 10 marks
1	A very weak response that: - includes little relevant knowledge and understanding of economic terminology, concepts and principles - includes application to the given context which, at best, is very weak - includes attempted analysis which is weak and unsupported.	1 – 5 marks

[25]

**10.****Areas for discussion include:**

- definition of imperfect information and market failure
- why imperfect information may be considered a source of market failure
- explanation of asymmetric information and how it can lead to some markets working inefficiently (or even collapsing altogether) when sellers know more about the true value of the good or service they are selling than the buyers (ie the 'Market for Lemons')
- explanation of moral hazard, where agents (knowing more about their risk profile than the insurer does) take excessive risk because they are protected from the negative consequences of their actions by insurance, thus leading to market failure
- how imperfect information can lead to the under-provision of merit goods and over-provision of demerit goods
- how imperfect information does not merely mean the absence of information. Information can also be imperfect if it is presented unhelpfully or incompletely such that making good decisions becomes very difficult
- agents may suffer from bounded rationality and cognitive biases
- agents may value short-term benefits too much due to a lack of information which could lead to poverty, eg lack of saving for pensions
- examples of markets in which imperfect information causes problems, eg the labour market, the cigarette market, the market for insurance.

The use of relevant diagrams to support the analysis should be taken into account when assessing the quality of the student's response to the question.

Level of response	Response	Max 15 marks
3	A good response provides an answer that: • is well organised and develops a selection of the key issues that are relevant to the question • shows sound knowledge and understanding of economic terminology, concepts and principles with few, if any, errors • includes good application of relevant economic principles to the given context and, where appropriate, good use of data to support the response • includes well-focused analysis with clear, logical chains of reasoning.	11 – 15 marks
2	A reasonable response provides an answer that: • focuses on issues that are relevant to the question • shows satisfactory knowledge and understanding of economic terminology, concepts and principles but some weaknesses may be present • includes reasonable application of relevant economic principles to the given context and, where appropriate, some use of data to support the response • includes some reasonable analysis but which might not be adequately developed or becomes confused in places.	6 – 10 marks
1	A weak response provides an answer that: • has identified one or more relevant issues • has some limited knowledge and understanding of economic terminology, concepts and principles but some errors are likely • has very limited application of relevant economic principles to the given context and/or data to the question • might have some limited analysis but it may lack focus and/or become confused.	1 – 5 marks

[15]